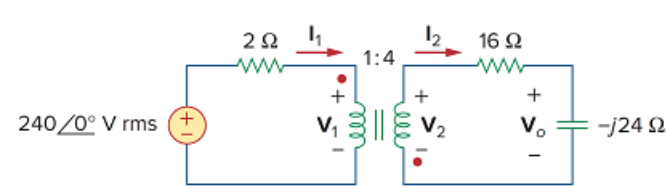


Problem

In the ideal transformer circuit of Fig. 13.38, find V_o and the complex power supplied by the source.

Figure 13.38



Step-by-step solution

Step 1 of 5

Draw the following circuit diagram:

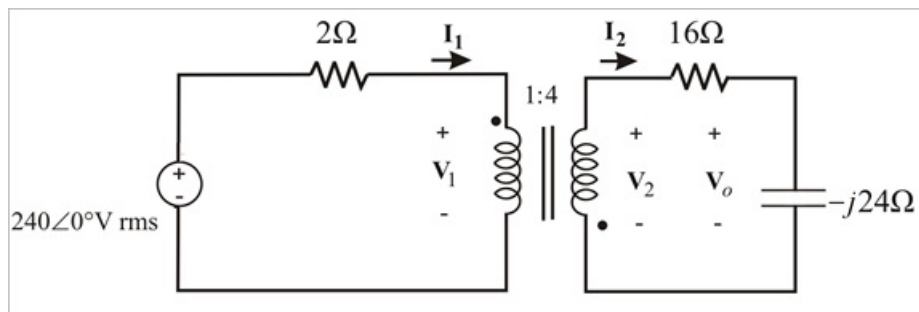


Figure 1

Step 2 of 5

Reflecting the secondary impedance $16 - j24 \Omega$ to primary side and calculate the value of the Z_R .

Write the expression for Z_R .

$$\begin{aligned} Z_R &= \frac{Z_s}{n^2} \\ &= \frac{16 - j24}{4^2} \\ &= 1 - j1.5 \Omega \end{aligned}$$

Calculate the value of the Primary current I_1 .

Write the expression for the Primary current I_1 .

$$\begin{aligned} I_1 &= \frac{V_{in}}{Z_1 + Z_R} \\ &= \frac{240 \angle 0^\circ}{2 + 1 - j1.5} \\ &= \frac{240 \angle 0^\circ}{3.54 \angle -26.56^\circ} \\ &= 71.55 \angle 26.56^\circ \end{aligned}$$

Therefore, the value of the Primary current I_1 is $71.55 \angle 26.56^\circ$ A.

Step 3 of 5

Calculate the value of the secondary current I_2 .

Since both currents are entering at dotted terminal write the expression for secondary current

I_2 .

$$\begin{aligned} I_2 &= -\frac{1}{n} I_1 \\ &= -\frac{1}{4} (71.55 \angle 26.56^\circ) \\ &= 17.88 \angle 206.56^\circ \end{aligned}$$

Therefore, the value of the secondary current I_2 is $17.88 \angle 206.56^\circ$ A.

Step 4 of 5

Calculate the value of the output voltage V_o .

Write the expression for output voltage V_o .

$$\begin{aligned} V_o &= (-j24) I_2 \\ &= (-j24) (17.88 \angle 206.56^\circ) \\ &= (24 \angle -90^\circ) (17.88 \angle 206.56^\circ) \\ &\cong 429.4 \angle 116.57^\circ \end{aligned}$$

Therefore, the value of the output voltage V_o is $429.4 \angle 116.57^\circ$ V

Step 5 of 5

Calculate the value of the Complex power S supplied by the source.

Write the expression for Complex power S supplied by the source.

$$\begin{aligned} S &= V_s I_1^* \\ &= (240 \angle 0^\circ) (71.55 \angle 26.56^\circ)^* \\ &= (240 \angle 0^\circ) (71.55 \angle -26.56^\circ) \\ &= 17.174 \angle -26.57^\circ \end{aligned}$$

Therefore, the value of the Complex power S supplied by the source is

$17.174 \angle -26.57^\circ$ kVA.